$He \longrightarrow He^{\oplus} + e$

$\boxed{He^{\oplus} \longrightarrow He^{+2} + e}$

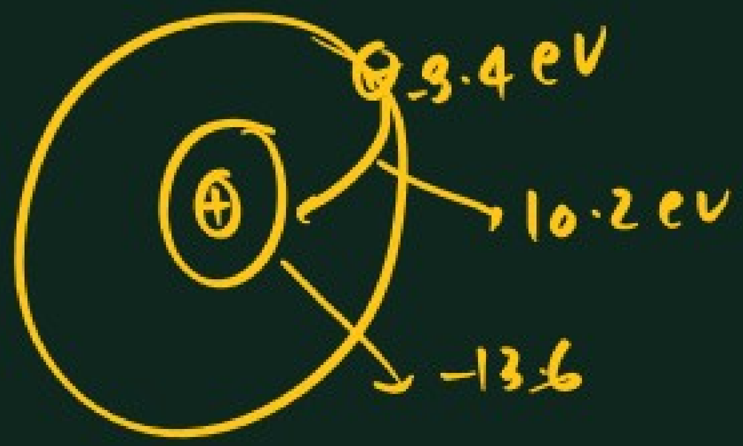
$He \longrightarrow He^{+2} + 2e$

$\varepsilon_1 = 24.5\,eV$

$\varepsilon_2 = 54.4$

$\varepsilon = ?$

$$E = -13.6 \times \frac{z^2}{n^2}$$



$$\Delta E = 13.6 \times z^2 \left(\frac{1}{1} - \frac{1}{\infty}\right)$$

$$= 13.6 \times 4$$

$$= 54.4$$

$n >>> 1$

$n+1 \simeq n$

$$mvr = \frac{nh}{2\pi}$$

$$\frac{2h}{\pi} = \frac{nh}{2\pi}$$

$\boxed{n_2 = 4}$

$n_1 = 3$

$$\frac{c}{\lambda} = cRz^2\left(\frac{1}{n^2} - \frac{1}{(n+1)^2}\right)$$

$$\nu = Rcz^2\left(\frac{\cancel{n^2} + 2n + 1 - \cancel{n^2}}{n^2(n+1)^2}\right) = Rcz^2\left(\frac{2n+1}{n^2(n+1)^2}\right) = \frac{Rcz^2(2n)}{n^4}$$

$\lambda = \frac{h}{mv}$

$\lambda = \frac{h}{\sqrt{2mK.E}}$

* If any particle acc' from rest by using potential V_0

$\boxed{KE = eV_0}$ e = charge

$$\boxed{\lambda = \frac{h}{\sqrt{2meV_0}}}$$

For $e^{\ominus} \rightarrow \lambda = \frac{12.27}{\sqrt{V_0}} = \sqrt{\frac{150}{V_0}}\ A^{\circ}$

(Proton) $P \rightarrow \lambda = \frac{0.286}{\sqrt{V_0}} A^{\circ}$

$\alpha \rightarrow$ Particle $\lambda = \frac{0.101}{\sqrt{V_0}} A^{\circ}$

Bohr's theory

→ particle nature

De-broglie

→ wave nature

no. of wave made by an $e^{\ominus}$ in any orbit = orbit no

$$\boxed{2\pi r = n\lambda}$$

$2\pi r = \frac{nh}{mv}$

$$\boxed{mvr = \frac{nh}{2\pi}}$$

1) $\lambda = \frac{h}{mv}$ Js

$m \rightarrow$ kg, $v \rightarrow$ m/sec

$= \frac{6.6\times10^{-34}}{66\times10^{-3}\times4\times10^{-2}} = \frac{66\times10^{-35}}{66\times4\times10^{-5}}$

$= 0.25\times10^{-30}$

$\lambda = 0.25\times10^{-30}$ m

2) $\lambda = \frac{h}{\sqrt{2mK.E}} = \frac{6.6\times10^{-34}}{\sqrt{2\times10^{-29}\times10^{-3}\times2\times10^{-19}}}$

$= \frac{6.6\times10^{-34}}{\sqrt{4\times10^{-46}}}$

$= \frac{6.6\times10^{-34}}{2\times10^{-23}}$

$= 3.3\times10^{-11}$ m

3) $\lambda = \frac{h}{mV}$

$m = 1.67\times10^{-27}$ kg

$v = 3\times10^{8}$ m/sec.

$\lambda = 1.3\times10^{-15}$ m

4) $2\pi r = n\lambda$ $\qquad r = 0.529\times4$

$2\pi r = 2\lambda$

$r\pi = \lambda$

$0.529\times4\times3.14 = \lambda$

$\lambda = 6.64$ A^{0}

$$\lambda_p = \frac{h}{m_p v_p}$$

$$\lambda_e = \frac{h}{m_e v_e}$$

$$\frac{\lambda_p}{\lambda_e} = \frac{1}{2}$$

$$\frac{\lambda_p}{\lambda_e} = \frac{\frac{\cancel{h}}{m_p v_p}}{\frac{\cancel{h}}{m_e v_e}}$$

$$\frac{\lambda_p}{\lambda_e} = \frac{m_e v_e}{m_p v_p}$$

$$\frac{v_p}{v_e} = \frac{\lambda_e}{\lambda_p} \times \frac{m_e}{m_p} = \frac{2}{1} \times \frac{\frac{1}{1840}}{1} = \frac{2}{1840} = \frac{1}{920}$$

$$\lambda_p = \frac{h}{m_p v_p}$$

$$\lambda_\alpha = \frac{h}{m_\alpha v_\alpha}$$

$$\frac{\lambda_p}{\lambda_\alpha} = \frac{\frac{1}{m_p v_p}}{\frac{1}{m_\alpha v_\alpha}}$$

$$\frac{\lambda_p}{\lambda_\alpha} = \frac{m_\alpha v_\alpha}{m_p v_p}$$

$$\frac{v_p}{v_\alpha} = \frac{m_\alpha}{m_p} \times \frac{\lambda_\alpha}{\lambda_p}$$

$$= \frac{4}{1} \times \frac{2}{1}$$

$$= \frac{8}{1}$$

$$\lambda_p = \frac{\cancel{h}}{\sqrt{\cancel{2} m_p K.E_p}}$$

$$\lambda_\alpha = \frac{\cancel{h}}{\sqrt{\cancel{2} m_\alpha K.E_\alpha}}$$

$$\frac{\lambda_p}{\lambda_\alpha} = \frac{\sqrt{m_\alpha K.E_\alpha}}{\sqrt{m_p K.E_p}}$$

$$\frac{1}{2} = \frac{\sqrt{m_\alpha K.E_\alpha}}{\sqrt{m_p K.E_p}}$$

$$\frac{1}{4} = \frac{m_\alpha}{m_p} \times \frac{K.E_\alpha}{K.E_p}$$

$$\frac{K.E_p}{K.E_\alpha} = \frac{16}{1}$$

$*$ $\lambda = \frac{12.27}{\sqrt{V_0}}$

$1.41 = \frac{12.27}{\sqrt{V_0}}$

$\sqrt{V_0} = \frac{12.27}{1.41}$

$V_0 = \frac{150}{2} = 75$

$1.73 = \frac{12.27}{\sqrt{V_0}}$

$3 = \frac{150}{V_0}$

$V_0 = 50$

$E_A = 4.25\,eV$ $\quad K.E_A$

$E_B = 4.2\,eV$ $\quad K.E_B$

$$\boxed{\begin{array}{l} \lambda_B = 2\lambda_A \\ K.E_B = K.E_A - 1.5 \quad \text{— (2)} \end{array}}$$

$K.E_A = E_A - (\varepsilon_0)_A$

$K.E_B = \varepsilon_B - (\varepsilon_0)_B$

$K.E_A = 2$

$K.E_B = 0.5$

$(\varepsilon_0)_A = 2.25$

$(\varepsilon_0)_B = 3.7$

$\lambda_A = \frac{h}{\sqrt{2mK.E_A}}$

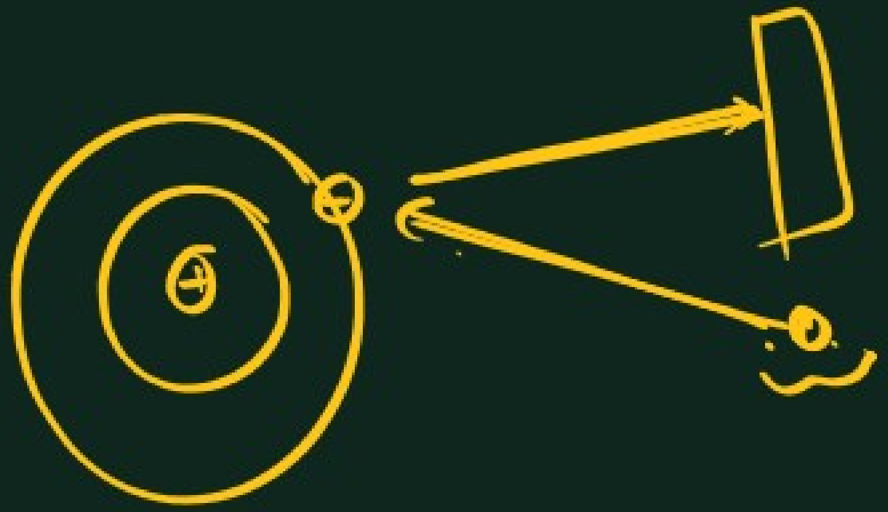
$\lambda_B = \frac{h}{\sqrt{2mK.E_B}}$

$\frac{\lambda_A}{\lambda_B} = \sqrt{\frac{K.E_B}{K.E_A}}$

$\frac{1}{4} = \frac{K.E_B}{K.E_A}$

$K.E_A = 4\,K.E_B$ — (1)

Heisenberg uncertainty principle →



$$\boxed{\Delta x \, \Delta p \geq \frac{h}{4\pi}}$$

Δx = uncertainty in position
Δp = -11- momentum

$$\Delta x \geq \frac{h}{4\pi \, \Delta p}$$

$$\Delta x \, \Delta p \geq \frac{h}{4\pi}$$

$$\Delta x \, \Delta mv \geq \frac{h}{4\pi}$$

$$\Delta x \, m \Delta v \geq \frac{h}{4\pi}$$

$$\boxed{\Delta x \, \Delta v \geq \frac{h}{4\pi m}}$$

$$\boxed{\Delta E \, \Delta t \geq \frac{h}{4\pi}}$$

Δv → uncertainty in velocity.

→ Applicable for sub-atomic Particle.

1) $\Delta x = \Delta p$

$\Delta x \, \underset{\sim}{\Delta p} \geq \frac{h}{4\pi}$

$\Delta x \, \Delta x \geq \frac{h}{4\pi}$

$(\Delta x)^2 = \frac{h}{4\pi}$

$\Delta x = \sqrt{\frac{h}{4\pi}}$

$\boxed{\Delta x = \frac{1}{2}\sqrt{\frac{h}{\pi}}}$

$\Delta x \, \Delta v \geq \frac{h}{4\pi m}$

$\frac{1}{2}\sqrt{\frac{h}{\pi}} \, \Delta v \geq \frac{h}{4\pi m}$

$\Delta v = \frac{1}{2m}\sqrt{\frac{h}{\pi}}$

2) $\Delta x \, \Delta v = \frac{h}{4\pi m}$

$\Delta v = \frac{h}{4\pi m \, \Delta x}$

$= \frac{6.6\times10^{-34}}{4\times3.14\times9\times10^{-31}\times0.33\times10^{-12}}$

$= \frac{\cancel{6.6}^{2}\times10^{-34}}{\cancel{4}^{2}\times\frac{22}{7}\times9\times10^{-31}\times\cancel{3.3}\times10^{-13}}$

$= \frac{7\times10^{-34}}{2\times22\times9\times10^{-44}}$

$= 1.7\times10^{8}$ m/sec.

3) $m = \frac{132}{\pi}$ gm

$\Delta v = 12.5\times10^{-24} \frac{m}{sec}$.

4) $2.2\times10^{6} \frac{m}{sec}$.

$\Delta v = \frac{0.1}{100}\times V$

$= \frac{0.1}{100}\times2.2\times10^{6}$

$= 2.2\times10^{3}$

$\Delta x = 2.6\times10^{-8}$ m

* Quantum Mechanical model →
→ Schrodinger wave eq^n →

$$\frac{d^2\psi}{dx^2} + \frac{d^2\psi}{dy^2} + \frac{d^2\psi}{dz^2} + \frac{8\pi^2 m}{h^2}(E-V)\psi = 0$$

ψ (Psi) → wave f^n of $e^{\ominus}$.
E → Total energy
V → P.E of $e^{\ominus}$
ψ → storehouse of $e^{\ominus}$.
→ Informat^n about $e^{\ominus}$.
→ ~~gau~~ provides → $\boxed{\text{quantum no.}}$
↓
$\boxed{n, l, m}$

Quantum no. →

1) Principal quantum no. (n)
2) Azimuthal quantum no. (l)
3) Magnetic quantum no. (m) ✓
4) Spin quantum no. (s)

1) Principal quantum no. → (n)

→ Represent size of orbital.

→ shell or orbit.

→ n takes the value from → 1 to ∞

→ max^m $e^\ominus = 2n^2$.

n	notat^n	max^m $e^\ominus = 2n^2$
1	K	2
2	L	8
3	M	18
4	N	32

2) Azimuthal quantum no. (l)

Angular momentum quantum no.

→ Represent subshells

→ l takes the value from → 0 to n-1

shell n	subshells l (0 to n-1)
1	0 ✓
2	0, 1
3	0, 1, 2
4	0, 1, 2, 3

l	not^n	max^m $e^\ominus$
0	s	2
1	p	6
2	d	10
3	f	14
4	g	18
5	h	22.

→ The no. of subshell in any shell = n

→ max^m no. of $e^\ominus$ in any subshell = $2(2l+1)$

3) Magnetic quantum no. (m)

→ Represent orientation of subshell in 3D space.

→ Represent → $\boxed{\text{orbital}}$ → Smallest region in which probability of finding $e^{\ominus}$ is max^m

→ m take the value from $-l$ to $+l$.

→ Each orbit contains max^m $\boxed{2e^{\ominus}}$

l	m $-l$ to $+l$
$0 \to s$	0
$1 \to p$	$-1, 0, +1$
$2 \to d$	$-2, -1, 0, +1, +2$
$3 \to f$	$-3, -2, -1, 0, +1, +2, +3$.

no. of orbitals in any subshell $= 2l+1$

4) spin quantum no. (s)

→ Represent spin of $e^{\ominus}$

→ Two $e^{\ominus}$ in any orbital always have opposite spin.

→ If 1st $e^{\ominus}$ → clockwise dir^n
2nd $e^{\ominus}$ → anticlockwise dir^n.

→ s take the value $\pm\frac{1}{2}$ for each m

n shell 1 to ∞	e⁻ $(2n^2)$	l subshell (0 to n-1)	e⁻ $2(2l+1)$	m orbital −l to +l	s $\pm\frac{1}{2}$ for each m
1 → K	2	0 → s	2	0	±1/2
2 → L	8	0 → s	2 ✓	0 ✓	±1/2
		1 → p	6 ✓	−1, 0, +1	±1/2, ±1/2, ±1/2
3 → M	18	0 → s	2 ✓	0	±1/2
		1 → p	6 ✓	−1, 0, +1	±1/2, ±1/2, ±1/2
		2 → d	10 ✓	−2, −1, 0, +1, +2	±1/2, ±1/2, ±1/2, ±1/2, ±1/2
4 → N	32	0 → s	2	0	±1/2
		1 → p	6	−1, 0, +1	±1/2, ±1/2, ±1/2
		2 → d	10	−2, −1, 0, +1, +2	±1/2, ±1/2, ±1/2, ±1/2, ±1/2
		3 → f	14	−3, −2, −1, 0, +1, +2, +3	±1/2, ±1/2, ±1/2, ±1/2, ±1/2, ±1/2, ±1/2

Q) How many sets of quantum no. are correct/possible →

	n	l	m	s
a)	3 ✓	✓ 1 ✓	-1	-1/2 ✓
b)	2	2	0	+1/2 ✗
c)	3 ✓	2 ✓	-3 ✗	-1/2
d)	4	3 ✓	-1	-1/2 ✓
e)	5	4 ✓	+2	+1/2 ✓
f)	6	4	+5 ✗	-1/2 ✗